INCLUSIONS OF THE ZARISKI AND ETALE COVERING CLASSES IN THE SMOOTH COVERING CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [1618, 1622] and proof(s) [[1624, 1629]]; this spec covers the complete environment instead (statement 1618-1622, proof span(s) [(1624, 1629)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. Definitional inclusion chain with cited facts.

Source reference. Original reference 02GI: sources/stacks-chow/tags.txt, source lines 3229–3229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0216: sources/stacks-chow/tags.txt, source lines 2659–2659. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Any étale covering is a smooth covering, and a fortiori, any Zariski covering is a smooth covering.*

Proof. This is clear from the definitions, the fact that an étale morphism is smooth see Morphisms, Definition 02GI and Lemma 0216. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.